

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 9: Triangles (Congruency in Triangles)

Topic-wise short notes with solved examples

The angle properties of a triangle, the conditions under which two triangles are congruent (SSS, SAS, ASA, AAS, RHS) and how to use congruent triangles to prove that sides or angles are equal.

1 Angles of a triangle

- The sum of the three angles of a triangle is 180° .
- An **exterior angle** equals the sum of the two **interior opposite angles**, so it is greater than either of them.
- A triangle can have at most one right angle or one obtuse angle.

Example 1.1 In triangle ABC, $\angle A = 50^\circ$ and the exterior angle at C is 120° . Find $\angle B$ and $\angle C$.

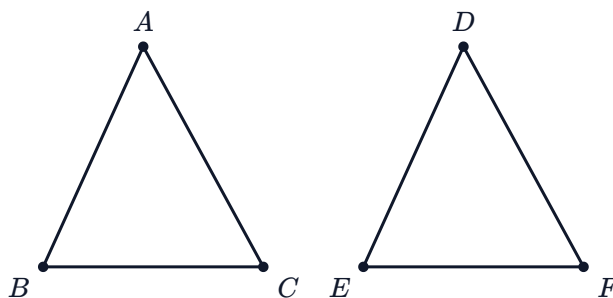
SOLUTION

$$\angle B = 120^\circ - 50^\circ = 70^\circ \text{ and } \angle C = 180^\circ - 120^\circ = 60^\circ$$

$$\angle B = 70^\circ, \angle C = 60^\circ$$

2 Congruent triangles

- Two triangles are **congruent** if they have exactly the same shape and size. One can be placed on the other so that it covers it exactly.
- We write $\triangle ABC \cong \triangle DEF$. The **order of the letters** shows which vertices correspond: A to D, B to E, C to F.
- C.P.C.T.**: corresponding parts of congruent triangles are equal. So $AB = DE$, $BC = EF$, $CA = FD$, $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$.



Triangle ABC is congruent to triangle DEF

3 Conditions for congruency

Condition	What must be equal
S.S.S.	Three sides of one triangle and the three sides of the other
S.A.S.	Two sides and the included angle
A.S.A.	Two angles and the included side
A.A.S.	Two angles and a corresponding side (not included)
R.H.S.	In right-angled triangles: the hypotenuse and one side

- **A.A.A. is not a condition:** equal angles give the same shape but not the same size.
- **S.S.A. is not a condition:** two sides and an angle that is not included do not fix the triangle (except in the R.H.S. case).

Example 3.1 In triangles ABC and PQR, $AB = PQ$, $\angle B = \angle Q$ and $BC = QR$. Are they congruent? If $\angle A = 65^\circ$, find $\angle P$.

SOLUTION

Two sides and the included angle are equal: S.A.S.

$\triangle ABC \cong \triangle PQR$, and $\angle P = \angle A = 65^\circ$ (C.P.C.T.)

Example 3.2 In triangles ABC and DEF, $\angle A = \angle D$, $\angle B = \angle E$ and $AB = EF$. Can we say they are congruent?

SOLUTION

AB lies between $\angle A$ and $\angle B$, but EF does not lie between $\angle D$ and $\angle E$. The equal sides are not corresponding.

No

4 Writing a proof

- State the two triangles.
- List three equal pairs, each with its **reason** (given, common side, vertically opposite angles, alternate angles, radii of the same circle, and so on).
- State the congruency with the correct condition and the vertices in corresponding order.
- Draw the conclusion by C.P.C.T.

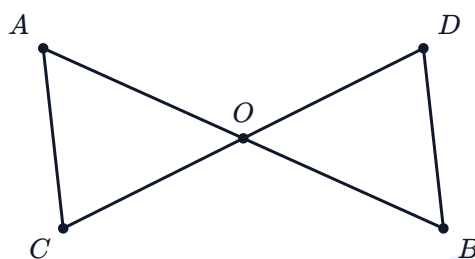
Example 4.1 AB and CD bisect each other at O. Prove that $AC = BD$ and that AC is parallel to BD.

SOLUTION

In $\triangle AOC$ and $\triangle BOD$: $AO = BO$ (given), $CO = DO$ (given), $\angle AOC = \angle BOD$ (vertically opposite).

So $\triangle AOC \cong \triangle BOD$ (S.A.S.), and $AC = BD$ (C.P.C.T.).

Also $\angle OAC = \angle OBD$ (C.P.C.T.). These are alternate angles, so $AC \parallel BD$.



AB and CD bisect each other at O

Example 4.2 In quadrilateral ABCD, $AB = AD$ and $CB = CD$. Prove that AC bisects $\angle BAD$.

SOLUTION

In $\triangle ABC$ and $\triangle ADC$: $AB = AD$, $CB = CD$, $AC = AC$ (common).

$\triangle ABC \cong \triangle ADC$ (S.S.S.)

$\angle BAC = \angle DAC$ (C.P.C.T.), so AC bisects $\angle BAD$

5 Results proved by congruency

- **Perpendicular bisector:** every point on the perpendicular bisector of a line segment is **equidistant from its two ends**. Conversely, a point equidistant from the ends lies on the perpendicular bisector.
- **Angle bisector:** every point on the bisector of an angle is **equidistant from the two arms**. Conversely, a point equidistant from the arms lies on the bisector.
- The perpendicular from the centre of a circle to a chord bisects the chord (R.H.S.).
- A diagonal of a parallelogram divides it into two congruent triangles (A.S.A.).

Example 5.1 P is a point on the bisector of $\angle ABC$. PM and PN are perpendiculars from P to BA and BC. Prove that $PM = PN$.

SOLUTION

In $\triangle PMB$ and $\triangle PNB$: $\angle PMB = \angle PNB = 90^\circ$, $\angle PBM = \angle PBN$ (BP bisects the angle), $BP = BP$ (common).

$\triangle PMB \cong \triangle PNB$ (A.A.S.), so $PM = PN$

Example 5.2 In right triangles ABC and DEF, $\angle B = \angle E = 90^\circ$, $AC = DF$ and $BC = EF$. If $AB = 5$ cm, find DE.

SOLUTION

Hypotenuse and one side are equal: R.H.S.

$\triangle ABC \cong \triangle DEF$, so $DE = AB = 5$ cm

6 Finding unknowns

- Once two triangles are known to be congruent, equate the corresponding parts to form equations.

Example 6.1 $\triangle ABC \cong \triangle PQR$. $AB = 3x + 1$, $PQ = 16$, $\angle C = 2y^\circ$ and $\angle R = 70^\circ$. Find x and y .

SOLUTION

$$3x + 1 = 16 \text{ and } 2y = 70$$

$$x = 5, y = 35$$

7 Common mistakes

- Writing the vertices in the wrong order, so that corresponding parts do not match.
- Using S.A.S. when the angle is not between the two sides.
- Claiming congruency from three equal angles.
- Using R.H.S. without a right angle, or with two sides that do not include the hypotenuse.
- Using C.P.C.T. before the triangles have been proved congruent.
- Leaving out the reason for a step.